

From paper plane to sci-fi racer

20 external visual references / 5 racing games / original low-poly implementation



A solid body. A readable cockpit. Two engines.

The new Aerolith craft uses a faceted central hull, independent engine pods, short stabilizers and a raised dark canopy. Its sci-fi character comes from thickness and component layout. The runtime model is 580 triangles, with no texture downloads.

Images in the research plates are official screenshots or concept art, credited beside each plate. Design takeaways are visual analysis of those images; they are not claims about the studios' internal modeling process. The cover is a render of the implemented Aerolith model.

What makes the silhouette work

A practical reading of the references, translated to this game

01 / Build volume

Give the nose, sides and rear their own planes. A visible underside and beveled shoulders stop the silhouette from reading as a thin triangular sheet.

02 / Separate functions

Use one cockpit, two engine masses and short connectors. Gaps between components carry the sci-fi character at a distance.

03 / Design for the chase view

The player mostly sees the rear. Frame bright engine cores with dark housings and keep the cockpit taller than the deck.

04 / Spend faces on outline

Add geometry where it changes the silhouette or catches light. Use flat materials for glass, chassis and stripes; omit tiny modeled vents and decals.

05 / Keep flight readable

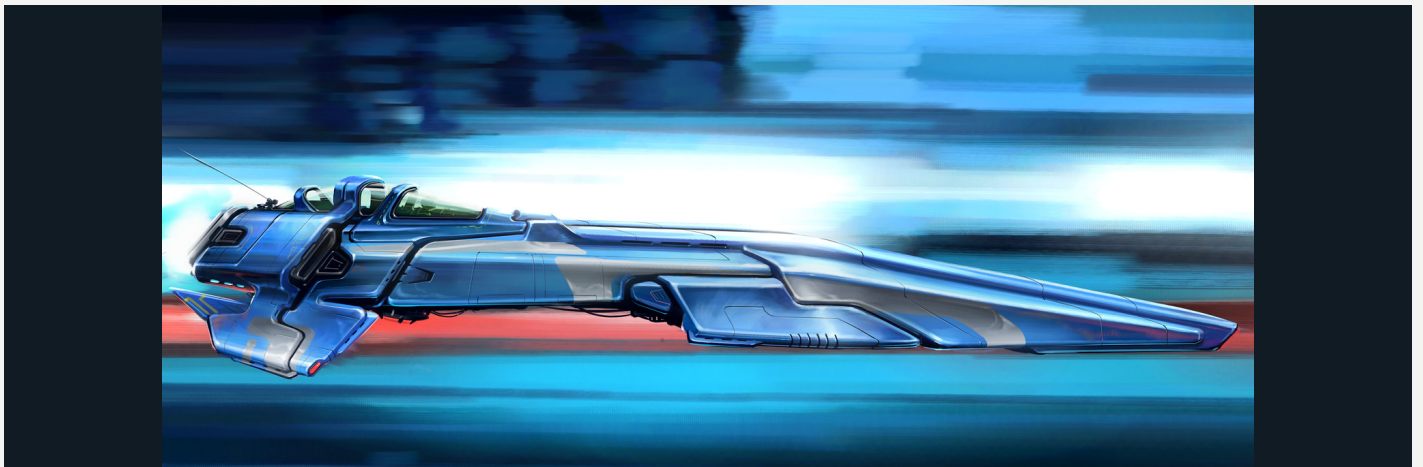
Preserve left/right symmetry and a clear forward axis. The short outer stabilizers stay near the existing wing-trail attachment points.

Chosen direction: a restrained twin-pod racer. BallisticNG is the closest reference for simple geometry; WipEout informs body/cockpit proportions; Redout 2 and FAST RMX inform engine separation and the rear signature; PACER informs visual mass.

WipEout

A racing machine with a human scale

These official concepts span several WipEout eras, rather than one uniform ship specification. They show a useful progression from recognizable cockpit and engine packaging to exaggerated racing proportions. The canopy establishes scale; deep engine housings and a continuous lower body make even a sharp nose feel manufactured. The blog includes artist commentary on cockpit interiors and race-grid staging. For Aerolith, borrow the separation of cockpit, chassis and propulsion, with broad surfaces that remain legible without decals. These are visual observations, not claims about the original polygon budgets.



01. Side profile: a long nose sits ahead of a substantial rear body. The lower edge and engine mass prevent the silhouette from reading as folded paper.



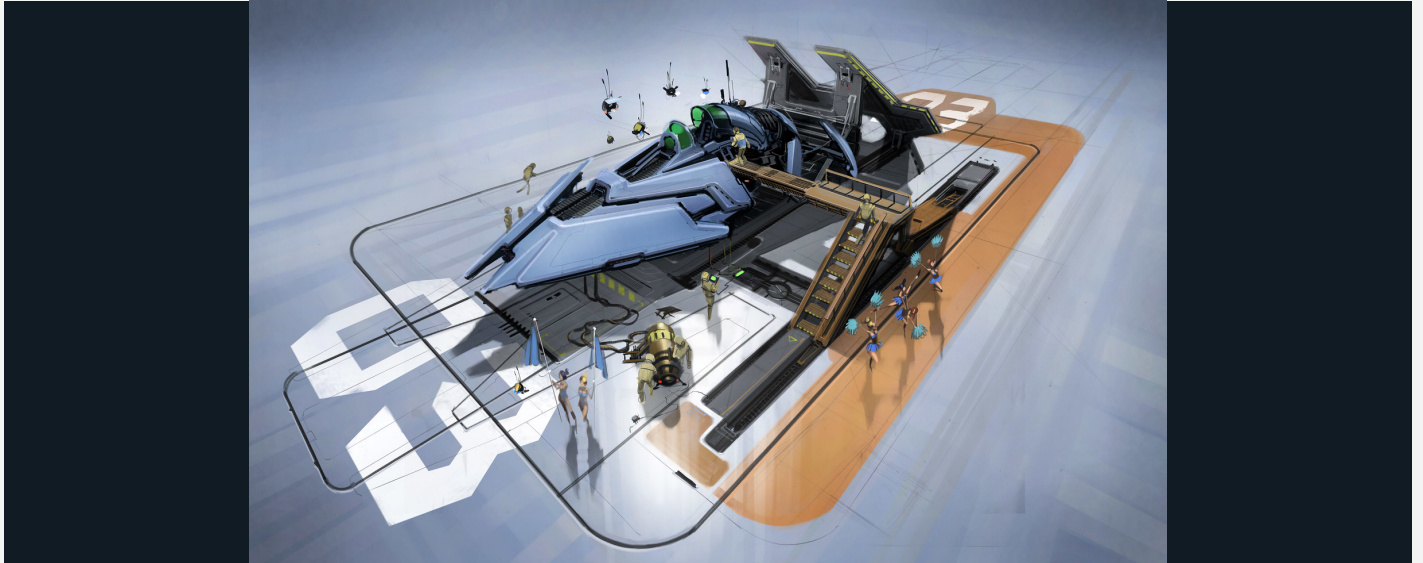
02. Cockpit cutaway: a seated pilot and canopy give the vehicle a believable scale. Aerolith can suggest this with one dark, raised canopy.

[Sony Interactive Entertainment / PlayStation Blog concept-art archive](#) - source

WipEout

A racing machine with a human scale / more angles

Two more views to test the design language beyond a single flattering angle.



03. Grid staging concept: separate armor sections, machinery and the central cockpit give the craft depth from an elevated camera.



04. Ground-level engine view: the propulsion unit is a major volume, not merely a colored mark on a flat wing.

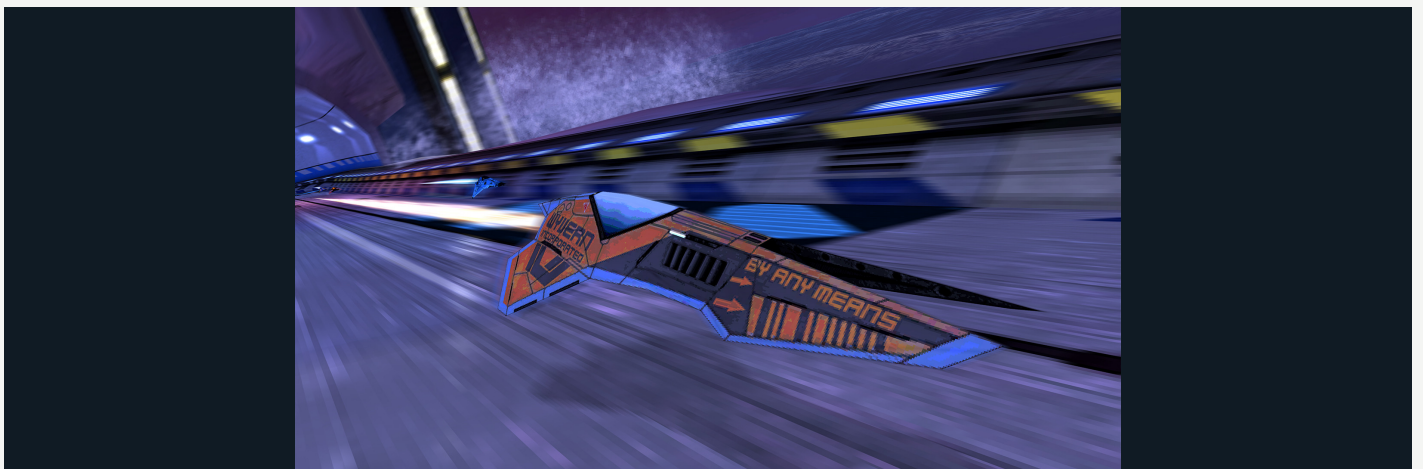
BallisticNG

The closest low-poly reference

Neognosis describes BallisticNG as a tribute to older anti-gravity racers and provides ship and livery modding tools. The screenshots demonstrate how much identity can come from a few angular volumes. Longitudinal color blocks, a dark canopy and clearly separated front and rear forms remain visible while banking. It is the most directly useful reference here for a deliberately low-poly result. Aerolith should retain those large shape changes and use actual geometry for its silhouette, while simplifying the dense text and team graphics.



05. A top-oblique view exposes separate body masses and negative space. A broad livery block reinforces the geometry rather than replacing it.



06. The angular canopy, long side surface and visible vent rhythm make the craft read as a machine even with coarse facets.

Neognosis / official [Steam store screenshots](#) - source

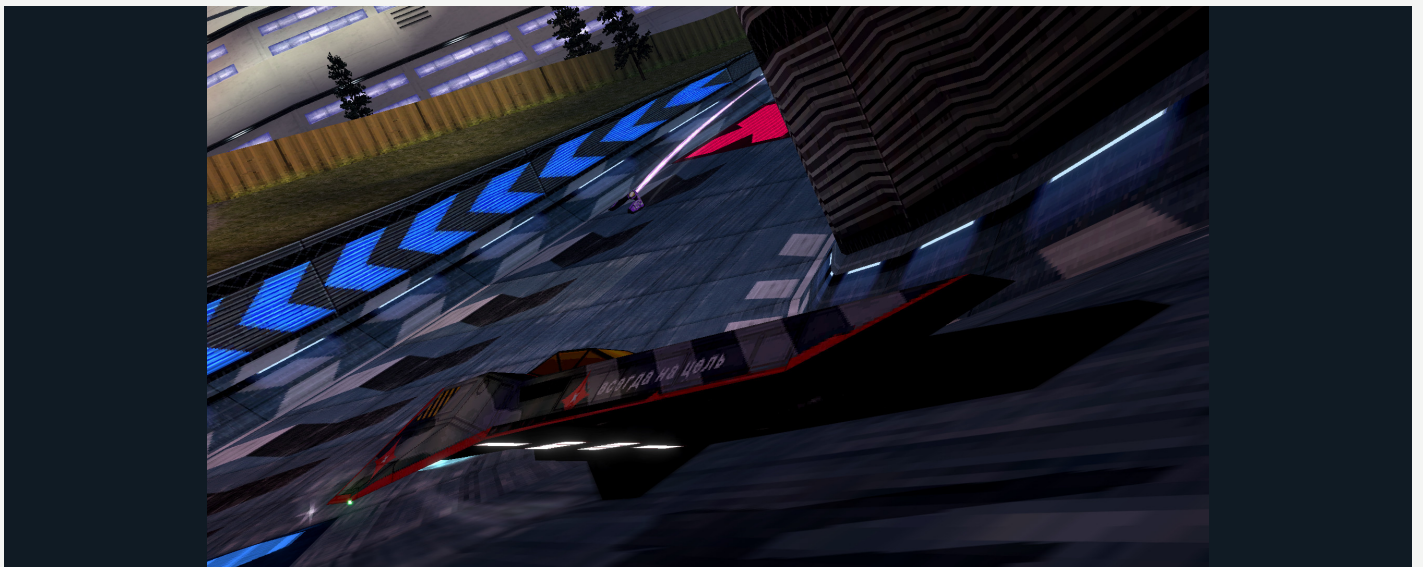
BallisticNG

The closest low-poly reference / more angles

Two more views to test the design language beyond a single flattering angle.



07. The same general design language remains readable at a steeper angle. Strong value separation does more work than tiny surface details.



08. The low side silhouette has a continuous mechanical body. A narrow profile can still have thickness, an identifiable cockpit and a definite tail.

Neognosis / official Steam store screenshots - source

Redout 2

Propulsion defines the rear silhouette

The official site describes 12 chassis and customization across propulsion, stabilizers, rudders, wings and other parts. In these screenshots, separated propulsion elements and pronounced appendages give the craft a mechanical identity from the chase camera. Bright exhaust establishes a clear rear-facing signature. Aerolith can borrow the two-engine rhythm and gaps between components, then reduce each component to a small number of planar faces. The reference is useful for packaging and rear-view readability; matching its surface complexity would work against the requested style.



09. Low rear three-quarter view: multiple engine outlets sit inside substantial housings. Light is anchored to a physical component.



10. A more unusual rear arrangement shows how propulsion placement changes the whole silhouette. Distinct engine spacing is recognizable without a logo.

34BigThings / official Steam store screenshots - source

Redout 2

Propulsion defines the rear silhouette / more angles

Two more views to test the design language beyond a single flattering angle.



11. The chase view emphasizes the cockpit, lateral components and exhaust. These are the features Aerolith needs to preserve at gameplay scale.



12. A wide rear stance and several bright outlets give immediate orientation. Use a restrained pair of outlets for Aerolith rather than this density.

PACER

Chunkier chassis, strong rear landmarks

The store screenshots show compact racing bodies with substantial rear surfaces and distinct cockpit placement. Their appeal comes partly from the relationship between a narrow forward section and a wider mechanical rear. In gameplay, that rear remains identifiable amid track markings, effects and other vehicles. Aerolith benefits from this same hierarchy: broad engine shoulders, a central cockpit and small stabilizers. Keep the engine signature visible, but leave weapons and combat effects out of the model brief.



13. Direct chase view: the broad rear surface and separated outlets are the dominant landmarks. This is the most important angle for the default camera.



14. A banked craft reveals a body with depth below its upper panels. The visible sides help communicate roll.

PACER

Chunkier chassis, strong rear landmarks / more angles

Two more views to test the design language beyond a single flattering angle.



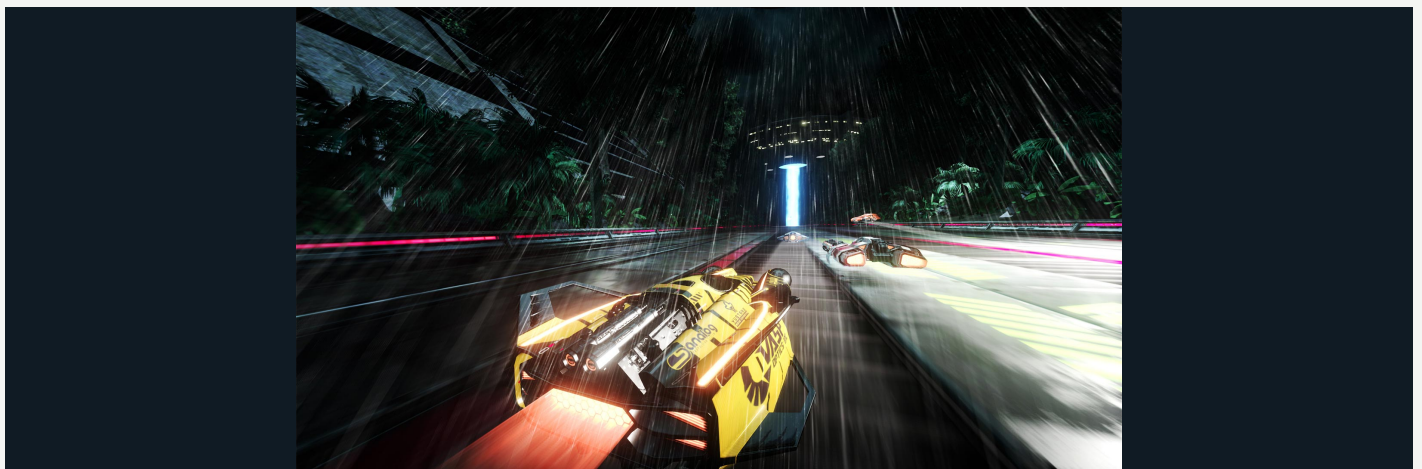
FAST RMX

Engine glow and clean shape families

The official page presents 15 racing machines and highlights boosting and phase switching. Its selection image is particularly useful because it shows a vehicle outside the visual noise of racing. Across the gameplay references, large luminous engine areas and simple body outlines remain prominent under motion blur. Aerolith can borrow the bright exhaust framed by a darker housing, plus a restrained lengthwise accent. Avoid letting engine glow erase the surrounding geometry; the ship must still work with post-processing disabled.



17. Vehicle-selection reference: the long chassis, elevated cockpit and broad engine housing form a readable shape before lighting effects are added.



18. Rear chase screenshot: a large illuminated outlet is enclosed by a solid shell. The contrast makes forward direction immediately obvious.

Shin'en Multimedia / official game-site images - source

FAST RMX

Engine glow and clean shape families / more angles

Two more views to test the design language beyond a single flattering angle.



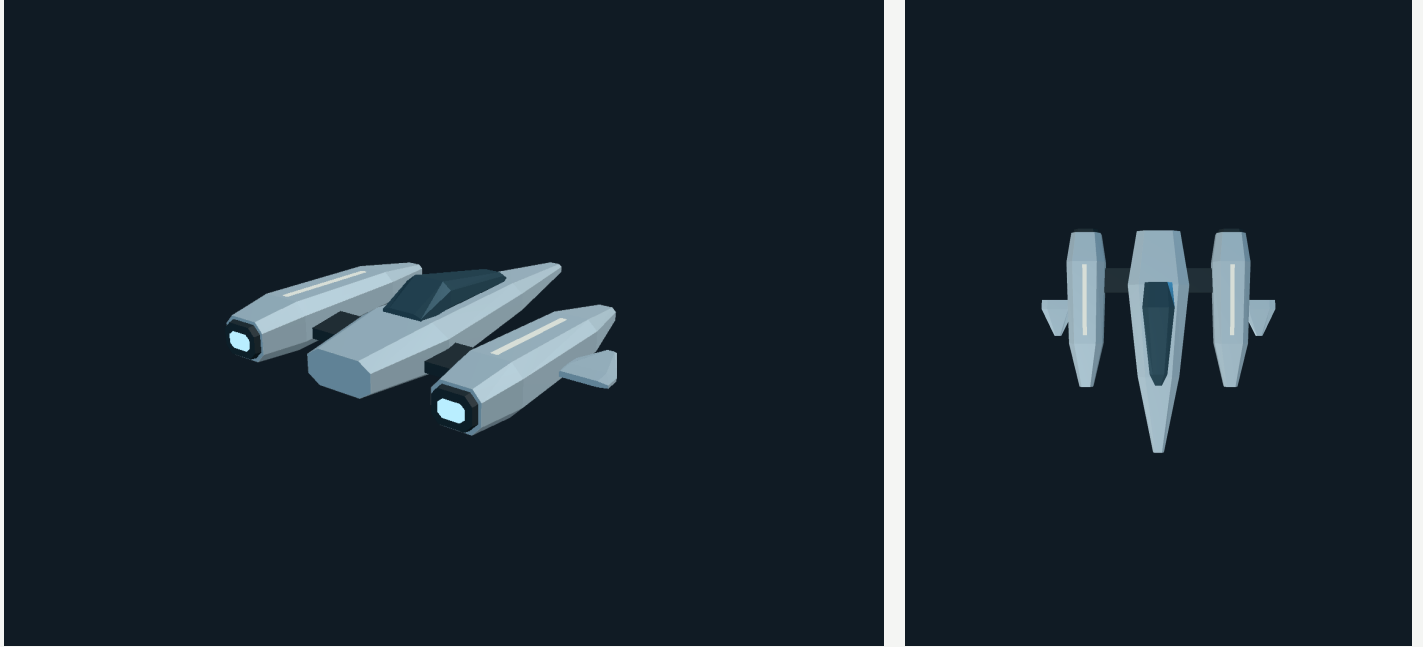
19. Banking screenshot: continuous side surfaces help the player read attitude even with substantial speed blur.



20. Two craft at different angles demonstrate why an exhaust landmark and a recognizable body outline need to work together.

Aerolith / Kestrel

Actual Three.js model, shown under neutral studio lighting



580 triangles / 15 mesh parts / 3 edge overlays

The pale armor is controlled by the existing Plane Color setting. The canopy, dark connectors and luminous engine cores retain their contrast. Eight-sided cross sections make broad bevels; none of the surfaces require smooth subdivision or external textures.

The overall envelope is about 11.4 wide, 10.8 long and 2.24 high in scene units. Two pods sit at $X = \pm 3.5$. The nose still points along +Z. Short stabilizers finish near the existing trail anchors, while the canopy provides the vertical break missing from the old model.

Why this reads differently: the original thin, broad delta wing dominated the outline. The replacement has three longitudinal masses, visible space between them, a raised canopy, and a substantial rear face with two engine outlets.

Model preview images are newly rendered from the repository implementation. They are not concept art copied from the referenced games.

The model is only part of the change

Pause, transparency, and evidence

White brackets now fade through transparency. Previously, each instance's RGB was multiplied by 0.08 + visibility x 0.92 while material opacity stayed at 0.94. That made the box nearly black before removal. A per-instance alpha attribute now reduces coverage while the chosen tint stays constant. Depth writing stays off; fog no longer darkens the overlay.

Pause freezes simulation, not rendering. The button and P / Esc freeze aircraft, flocks, wind, trails, probe expansion/decay, route fade and CRT time. Pending terrain-worker results wait until resume. Orbit, camera presets and appearance controls remain live. Flight-mode changes, reset and new probe triggers cannot advance the paused scene. Engine and wind audio fade to silence.

Validation. Automated browser checks compare world state across paused frames while confirming rendered frame counts increase. Camera selection, orbit, live color edits, resume, shortcuts and the Ambient-mode button are exercised. Unit regression tests cover frozen camera/FOV, paused orbit, and box alpha/tint. Type checking, the full test suite and the production build are also checked.

Primary sources and image credits

[WipEout](#) / [Sony Interactive Entertainment](#) / [PlayStation Blog](#) concept-art archive

[BallisticNG](#) / [Neognosis](#) / [official Steam store](#) screenshots

[Redout 2](#) / [34BigThings](#) / [official Steam store](#) screenshots

[PACER](#) / [R8 Games](#) / [official Steam store](#) screenshots

[FAST RMX](#) / [Shin'en Multimedia](#) / [official game-site](#) images

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